

Multimodal AI for Product Understanding

Problem Statement

E-commerce platforms like Amazon and Flipkart receive thousands of new product listings every day. Sellers upload a photo, but the platform still needs to know what the product actually is in order to categorize it, write a description, and make it searchable. This assignment designs a system that looks at a single product image and automatically understands what the product is.

1. Inputs and 2. Outputs

Input	Output
A single product image (e.g. a photo uploaded by the seller)	• Product category (e.g. “Footwear → Running Shoes”) • Product description (a short, natural-language summary)

3. Model Selection

We need a model that understands both images and text together — this is called a multimodal (vision-language) model.

- CLIP (Contrastive Language–Image Pretraining): learns a shared space where matching images and text descriptions land close together. Great for classifying a product into a category and for visual search.
- BLIP / BLIP-2: built for image captioning — it can generate a natural-language sentence describing what is in the image, which is exactly what we need for “product description.”

In practice, the two are complementary: CLIP is strong at categorization and similarity matching, while BLIP is strong at generating descriptive text. A production system can use both together.

4. System Design (High-Level)

1. Seller uploads a product image.
2. Preprocessing: resize/normalize the image so it matches what the model expects.
3. Category classification: pass the image through CLIP’s image encoder and compare it against text embeddings of known categories (e.g. “shoe”, “watch”, “shirt”) to find the closest match.
4. Description generation: pass the same image through BLIP to generate a natural-language caption.
5. Post-processing: combine the predicted category and generated description into structured product metadata.
6. Store and index: save the metadata (and the image embedding) so the product can appear in search and recommendations.

Image → CLIP (category) + BLIP (description) → Combined metadata → Catalog / Search index

5. Evaluation

- Accuracy: compare predicted categories against a labeled test set of products and measure what percentage are classified correctly.

- Quality of generated descriptions: use automatic text-similarity scores (like BLEU or CLIPScore, which checks if the description actually matches the image) and spot-check a sample of outputs manually.
- Human feedback: track how often sellers edit or reject the auto-generated category/description — frequent edits signal the model needs improvement.

6. Business Value

- Faster seller onboarding: sellers do not need to manually fill out category and description fields for every product.
- Better search and discovery: accurate categories and descriptions make products easier for shoppers to find.
- Consistency at scale: thousands of listings get uniformly tagged, instead of relying on inconsistent manual entry.
- Lower operational cost: less manual catalog review and correction work for the platform’s operations team.

Bonus: Real-World Example — Pinterest

Pinterest uses multimodal AI to power its visual search and “Lens” features. It combines Visual Language Models with multimodal embeddings to understand an image’s style, color palette, and product type, then matches it to shoppable products and visually similar pins. Pinterest has reported that this unified, multimodal approach to search noticeably improved relevance and engagement compared to older, text-only search.